

Curve Sketching.

Even function:- If $f(-x) = f(x)$
↓
Eg:- $\cos x, x^2$, etc.

Symmetric about y-axis

Odd function:- If $f(-x) = -f(x)$
↓
Eg:- $x^3 + x, \tan x$, etc.

Symmetric about origin

Periodicity: Repeating of graph pattern in certain interval

To test periodicity,

① Find period (k) by

$$k = \frac{\alpha}{|a|}$$

$\alpha \rightarrow$ Fundamental period

$a \rightarrow$ Constant term of angle

① Show, $f(x) = f(x+k)$

Fundamental period: (i) Of $\sin x$ & $\cos x$ is 2π

(ii) Of $\tan x$ is π .

Increasing function: A function $f(x)$ is said to be increasing in an interval (a, b) if for all $x_1, x_2 \in (a, b)$

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$$

Decreasing function: A function $f(x)$ is said to be decreasing in an interval (a, b) if for all $x_1, x_2 \in (a, b)$

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$$

$$x_1, x_2 \in \text{given interval}$$

$$x_1, x_2 \in \text{given interval}$$

Characteristics for sketching the curve of the function:-

(i) Origin [Passes or not] [By keeping $x=0$, if $y=0$; pass]

(ii) Domain & Range [x & y -axes ma kati space likh)

(iii) Symmetry [Even function - Symmetric about y-axis
Odd function - Symmetric about origin]

(iv) Periodicity [For trigonometric function]

(v) Asymptote: [For Logarithmic, exponential, tangent]

↳ [If $x=a, y \rightarrow \infty$ (vertical asymptote)]

[If $y=a, x \rightarrow \infty$ (horizontal asymptote)]

(vi) Increasing and decreasing region.

(vii) Openness [Quadratic, $ax^2+bx+c=0$ $a < 0$ opens downward
 $a > 0$ opens upward]

(viii) Vertex [For quadratic, $(h, k) = (-\frac{b}{2a}, \frac{4ac-b^2}{4a})$]

(ix) points on axes

Quadratic function

Characteristics:-

- (i) Openness (ii) Origin (iii) Vertex (h, k) (iv) line of symmetry $x = h$
- (v) Cutting x -axis ($at y=0$) (vi) Domain = $(-\infty, \infty)$
Range = $[k, \infty)$ if $a > 0$
 $(-\infty, k]$ if $a < 0$.

For cubic function:-

- (i) Origin (ii) Symmetry (iii) Increasing and decreasing parts.
- (iv) Points on axes. (v) Domain = $(-\infty, \infty)$
Range = $(-\infty, \infty)$

For sine and cosine function:-

- (i) Origin (ii) Amplitude [$y = b \sin ax$ or $y = b \cos ax$
 $b \rightarrow$ amplitude]
- (iii) Point on axes (iv) Period $(T) = \frac{2\pi}{a}$ (v) Domain $(-\infty, \infty)$ or given interval
Range $[-b, b]$
- (vi) Symmetry [Odd $\rightarrow$ origin Even $\rightarrow y$ -axis]

For tangent function [$y = a \tan bx$]

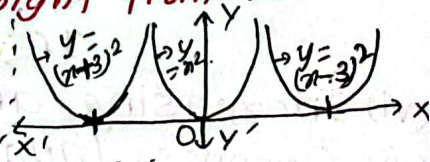
- (i) Origin (ii) Domain = $(-\infty, \infty)$ or given interval; Range = $(-\infty, \infty)$
- (iii) Period $(T) = \frac{\pi}{a}$ (iv) Symmetry (origin)
- (v) Points on axes (vi) Asymptote.

For exponential function [$y = \log_a x \Rightarrow x = a^y$]

- (i) Origin (ii) Asymptote (iii) Points where x +ve / x -ve
lie above/below x/y axes.
- (iv) Points on axes (v) Cuts x -axis; y -axis
- (vi) Increasing & decreasing.

Horizontal shifting of curve:- The graph of $f(x-a)$ is same as the graph of $f(x)$ moved to 'a' units right from the graph of $y = f(x)$. Note,

$y = f(x-a)$, ($a > 0$), shifted 'a' units right
 $y = f(x+a)$, ($a > 0$), shifted 'a' units left



Vertical shifting of curve:-

- (i) Graph of $y = f(x) + b$ ($b > 0$) is graph of $y = f(x)$ shifted b units upward.
- (ii) Graph of $y = f(x) - b$ ($b > 0$) is graph of $y = f(x)$ shifted b units downward.

Focal distance = $x+a$ } For horizontal parabola
 Focal distance = $y+a$ } For vertical parabola.

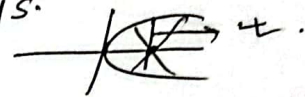
For vertex $\rightarrow$ Partial derivative (w.r.t x & w.r.t y)

Axis $\rightarrow$ Partial derivative

• Length of latus rectum $\Rightarrow y^2 \Rightarrow x$ ko coeff. { keeping coeff. of y^2 1 }
 $\Rightarrow x^2 \Rightarrow y$ ko coeff.

• Vertex is shortest distance from focus.

• Latus rectum is shortest focal chord



• If $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ are extremities of focal chord then $t_1 t_2 = -1$

& If these point subtend rt. angle at vertex then,

$$t_1 t_2 = -4$$

• Locus of mid-point of chord to $y^2 = 4ax$ passing through vertex $\rightarrow y^2 = 2ax$ $\leftarrow \frac{1}{2}$

• $y^2 = x \Rightarrow$ even in $y \rightarrow$ symmetrical about x -axis
 $\Rightarrow$ even in x $\rightarrow$ y -axis
 $\Rightarrow$ even in both $\rightarrow$ both axes.
 { Circle, ellipse, hyperbola }

✓ Tangent at vertex, linear term = 0.

• For tangent, $x^2 = 4ay$

Take line to this form $\rightarrow x = my + c$
 $\hookrightarrow c = a/m$

} Interchanging method.

• Parabola $\rightarrow$ Quadratic eqⁿ not linear.

• Slope = $\frac{dy}{dx}$ } of tangent

Slope = $-\frac{1}{dy/dx}$ } Normal

✓ Angle b/w tangent from (x_1, y_1) to $y^2 = 4ax$

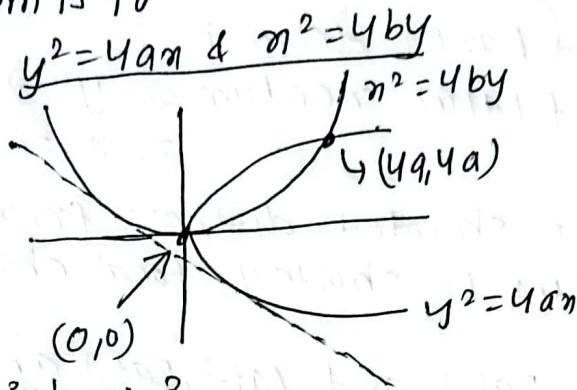
$\hookrightarrow \tan^{-1} \left(\frac{\sqrt{y_1^2 - 4ax_1}}{x_1 + a} \right)$ Root of S_1
 Direction.

$\tan^{-1} \frac{NS}{Directrix}$

- Angle b/w tangents at end points of focal chord $\Rightarrow 90^\circ$ {Meet at directrix}
- If tangents drawn from parabola meet at directrix then Angle b/w them is 90°

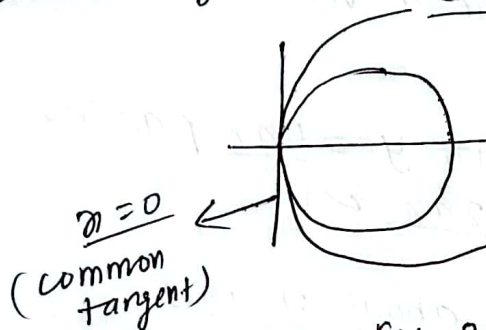
- Common tangent b/w $y^2 = 4ax$ & $x^2 = 4by$

$$a^{1/3}x + b^{1/3}y + a^{2/3}b^{2/3} = 0$$



- Max. normal from a point $\rightarrow 3$

- Common tangent b/w $(x-2a)^2 + y^2 = (2a)^2$ & $y^2 = 4ax$



circle

- If normal at point $(at_1^2, 2at_1)$ meet at another point $(at_2^2, 2at_2)$ on parabola $y^2 = 4ax$, then $t_2 = -t_1 - \frac{2}{t_1}$

- Condition of tangency of line $y = mx + c$ to $y^2 = 4ax \Rightarrow c = a/m$

- Condition of tangency of $y = mx + c$ to $y^2 = 4a(x+a)$

$$y = m(x+a) + (c-am)$$

$$\Rightarrow (c-am) = \frac{a}{m}$$

$$\Rightarrow c = am + \frac{a}{m}$$

- Eqⁿ of normal of parabola $y^2 = 4ax$ having slope 'm'

$$\hookrightarrow y = mx - 2am - am^3$$

- & POC is $(am^2, -2am)$ } $m \rightarrow$ slope of normal.

- POC $\Rightarrow$ Total

- Eqⁿ of given parametric form } Anyhow Parameter $\Rightarrow$ 2

- No. of tangents } Given point (x_1, y_1) $\begin{cases} \text{outside} \rightarrow 2 \\ \text{on} \rightarrow 1 \\ \text{inside} \rightarrow 0 \end{cases}$ } $\begin{cases} y_1^2 - 4ax_1 > 0 \\ y_1^2 - 4ax_1 = 0 \\ y_1^2 - 4ax_1 < 0 \end{cases}$

- The polar of (x_1, y_1) w.r.t. $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$ $\leftarrow$ Tangent to Circle at the point

→ $\log_e K \Rightarrow$ is not defined for $K \leq 0$

→ No. of solutions: No. of points where graphs cut each other

$f(x) = g(x) \Rightarrow$ curves are $y = f(x)$ & $y = g(x)$

→ ${}^nC_r = {}^nC_s \Rightarrow r = s$ or $r + s = n$

~~${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$~~

~~$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$~~

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

Selecting one or more object out of $n = 2^n - 1$.

$\Rightarrow$ If only two ways for each object

$${}^nP_r = \frac{n!}{(n-r)!} \cdot {}^nC_r$$

Derivatives

Some useful derivatives:

$$(1) \frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$$

$$(2) \frac{d}{dx} x^n = n x^{n-1} \left[\text{Eg: } \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}} \quad \& \quad \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2} \right]$$

$$(3) \frac{d}{dx} [f(x)]^n = n [f(x)]^{n-1} f'(x)$$

$$(4) \frac{d}{dx} e^{f(x)} = e^{f(x)} \cdot f'(x)$$

$$(5) \frac{d}{dx} a^{f(x)} = a^{f(x)} \cdot f'(x) \log a$$

$$(6) \frac{d}{dx} \log f(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$(7) \frac{d}{dx} \log x = \frac{1}{x}, \text{ for } x > 0$$

$$(8) \frac{d}{dx} \log |x| = \frac{1}{x}, \text{ for } x \neq 0$$

$$(9) \frac{d}{dx} (\log_a x) = \frac{d}{dx} \left(\frac{\log x}{\log a} \right) = \frac{1}{x \log a}, \text{ for } x > 0$$

$$(10) \frac{d}{dx} |x| = \frac{x}{|x|} \text{ or } \frac{|x|}{x}$$

Derivatives of trigonometric functions:

$$(1) \frac{d}{dx} \sin x = \cos x \quad (2) \frac{d}{dx} \cos x = -\sin x \quad (3) \frac{d}{dx} \tan x = \sec^2 x$$

$$(4) \frac{d}{dx} \sec x = \sec x \tan x \quad (5) \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x \quad (6) \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

Derivatives of inverse trigonometric functions:

$$(1) \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$(2) \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}, \quad \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$(3) \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}, \quad \frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$$

Derivative of hyperbolic functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

note

$$1. \frac{d}{dx} \sinh x = \cosh x$$

$$2. \frac{d}{dx} \cosh x = \sinh x$$

$$3. \frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

$$4. \frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$

$$5. \frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \coth x$$

$$6. \frac{d}{dx} \coth x = -\operatorname{cosech}^2 x$$

$$\# f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

↳ provided that this limit exists and is finite

$$\# \textcircled{i} \frac{d}{dx} (u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

$$\textcircled{ii} \frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\textcircled{iii} \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\textcircled{iv} \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \quad \left. \begin{array}{l} \text{If } y = f(t) \\ t = f(x) \end{array} \right\}$$

Derivative of parametric functions
If $x = \phi(t)$ and $y = \psi(t)$, then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Partial derivative:

For $f(x, y) = 0$

$$\frac{dy}{dx} = - \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

For u-substitution

$$\sqrt{\boxed{u}}$$

$$e^{\boxed{u}}$$

$$\frac{1}{\boxed{u}}$$

$$\boxed{u}^{\boxed{x}}$$

$$\frac{1}{\sqrt{\boxed{u}}}$$

$$\frac{1}{(\boxed{u})^2}$$

$$u) + C$$

$$\log \boxed{u}$$

$$\sin \boxed{u}, \cos \boxed{u}$$

Formulae:-

Integration (Anti-Derivatives)

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c \quad [n \neq -1]$$

$$2. \int e^x dx = e^x + c$$

$$3. \int e^{ax} dx = \frac{e^{ax}}{a} + c$$

$$4. \int \frac{1}{x} dx = \log x + c$$

$$5. \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c$$

$$6. \int \frac{1}{ax+b} dx = \frac{1}{a} \log(ax+b) + c$$

$$7. \int a^x dx = \frac{a^x}{\log a} + c$$

$$8. \int \sin x dx = -\cos x + c$$

$$9. \int \sin ax dx = -\frac{\cos ax}{a} + c$$

$$10. \int \cos x dx = \sin x + c$$

$$11. \int \tan x dx = -\log |\sec x| + c$$

$$12. \int \sec^2 x dx = \tan x + c$$

$$13. \int \sec^2 ax dx = \frac{\tan ax}{a} + c$$

$$14. \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$15. \int \sec x \cdot \tan x dx = \sec x + c$$

$$16. \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$17. \int \tan x dx = \log |\sec x| + c$$

$$18. \int \cot x dx = \log |\sin x| + c$$

$$19. \int \sec x dx = \log |\sec x + \tan x| + c$$

$$20. \int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + c$$

$$\log_e a^x = \frac{\log a^x}{\log e} = \frac{x \log a}{1} = x \log a$$

$$a^{\log_a b} = b$$

$$21. \int 1 dx = x + c$$

$$22. \int e^{(ax+b)} dx = \frac{1}{a} e^{(ax+b)} + c$$

$$23. \int k f(x) dx = k \int f(x) dx$$

$$24. \int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$$25. \int (uv) dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx + C, \text{ this}$$

is known as
integration by
parts:

$u \rightarrow$ easily derivable
 $v \rightarrow$ easily integrable

26. For, (i) $a^2 - x^2$ integrable
 $x = a \sin \theta$ $\hookrightarrow$ ILATE rule.

(ii) $x^2 - a^2$
 $x = a \sec \theta$

(iii) $x^2 + a^2$
 $x = a \tan \theta$

For u-substitution

$$\sqrt{u} \quad e^u \quad \frac{1}{u}$$

$$\frac{1}{\sqrt{u}} \quad \frac{1}{u^2}$$

$$\log u \quad \sin u \quad \cos u$$

Integration by parts :-

$$\int u v dx = u \int v dx - \int \left[\frac{du}{dx} \int v dx \right] dx$$

$u \rightarrow 1^{st} \text{ func}^n$
 $v \rightarrow 2^{nd} \text{ function}$

$u \rightarrow \text{easily derivable}$
 $v \rightarrow \text{easily integrable}$

order of choosing 1st f(x)

I = Inverse function
 L = Logarithmic function
 A = Algebraic function
 T = Trigo (function)
 E = Expo. function

* If inverse function & logarithmic function

has no second product, then take 'I' as second product.

* If one function is x^n , $n \in \mathbb{N}$, then

$$\int x^n v dx = x^n \int v dx - \frac{F(x)}{n}$$

$\rightarrow F(x)$ is obtained by applying the rule $\frac{(\text{derivative of I}) \times (\text{Integration of II})}{n}$

on every previous term obtained & writing them in alternate sign.

eg: $\int x^3 e^{ax} dx = \underline{\underline{\frac{x^3}{a} e^{ax}}} - \underline{\underline{\frac{3x^2}{a^2} e^{ax}}} + \underline{\underline{\frac{6x}{a^3} e^{ax}}} - \underline{\underline{\frac{6}{a^4} e^{ax}}} + C$

Integral of type $\int \sin^m x \cos^n x dx$

$\hookrightarrow$ (i) m -odd $\Rightarrow \cos x = t$

(ii) n -odd $\Rightarrow \sin x = t$

उत्तर को power odd xing करतार 't' put garna.

Integrals of type $\int \sin^m x \cos^n x dx$, $\int \sin^m x \sin^n x dx$, $\int \cos^m x \cos^n x dx$

Use trigonometric identities.

- (i) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$
- (ii) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$
- (iii) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$
- (iv) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

Standard integrals of Trigo.

$$(a) \int \operatorname{cosec} x dx = \log \left(\tan \frac{x}{2} \right) + c$$

$$(b) \int \sec x dx = \log \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right) + c.$$

★ Anyhow try to make $\sec^2$ in numerator and $\tan^2$ or $\tan$ in denominator.

• linear term in deno:

$$\text{eg: } \int \frac{1}{a+b \sin x} dx, \int \frac{1}{a \sin x + b \cos x} dx, \int \frac{dx}{a+b \cos x}$$

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}, \cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$\Rightarrow \tan \frac{x}{2} = t \text{ then } 2dt = \sec^2 \frac{x}{2} dx$$

• square term in deno. (degree 2)

$$\text{eg: } \int \frac{1}{a+b \sin^2 x}, \int \frac{1}{a \sin^2 x + b \cos^2 x} dx$$

• divide num & deno by $\cos^2 x$ and put $\tan x = t \Rightarrow dt = \sec^2 x dx$.

Next, Use Standard integrals (I).

For integral of the form

$$\int \frac{dx}{a \sin x \pm b \cos x}$$

$b \rightarrow \cos$ $\frac{a}{\text{coeff}}$
 $a \rightarrow \sin$ $\frac{b}{\text{coeff}}$

$$\text{Putting } a = r \cos \theta, b = r \sin \theta, r = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \left(\frac{b}{a} \right)$$

$$I = \int \frac{dx}{r \sin(x \pm \theta)} = \frac{1}{\sqrt{a^2 + b^2}} \ln \tan \left(\frac{x}{2} \pm \frac{1}{2} \tan^{-1} \frac{b}{a} \right) + c$$

$$\# \int a^x dx = \frac{a^x}{\log a} + c \quad \# \int e^{ax} dx = \frac{e^{ax}}{a} + c$$

$$(ii) \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c = -\cos^{-1} x + c$$

$$(iii) \int \frac{dx}{1+x^2} = \tan^{-1} x + c = -\cot^{-1} x + c$$

$$(iv) \int \frac{dx}{x \sqrt{x^2-1}} = \sec^{-1} x + c = -\operatorname{cosec}^{-1} x + c$$

$$\# \int \tan x dx = \ln \sec x + c = -\ln \cos x + c$$

$$(ii) \int \cot x dx = \ln \sin x + c$$

$$(iii) \int \sec x dx = \log (\sec x + \tan x) + c = \log \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) + c$$

$$(iv) \int \operatorname{cosec} x dx = \log (\operatorname{cosec} x - \cot x) + c = \log \left(\tan \frac{x}{2} \right) + c$$

Hyperbolic function

$$(i) \int \sinh x dx = \cosh x + c$$

$$(ii) \int \cosh x dx = \sinh x + c$$

$$(iii) \int \tanh x dx = \log (\cosh x) + c$$

$$(iv) \int \coth x dx = \log (\sinh x) + c$$

$$(v) \int \operatorname{cosech} x dx = \log \tan \left(\frac{x}{2} \right) + c$$

$$(vi) \int \operatorname{sech} x dx = 2 \tan^{-1} \left(\tanh \frac{x}{2} \right) + c$$

$$(vii) \int \operatorname{sech}^2 x dx = \tanh x + c$$

$$(viii) \int \operatorname{cosec}^2 x dx = -\coth x + c$$

$$(ix) \int \operatorname{sech} x \tanh x dx = -\operatorname{sech} x + c$$

$$(x) \int \operatorname{cosech} x \coth x dx = -\operatorname{cosech} x + c$$

$$\# \sinh x = \frac{e^x - e^{-x}}{2}; \cosh x = \frac{e^x + e^{-x}}{2}$$

$$(i) \cosh^2 x - \sinh^2 x = 1$$

$$(ii) \cosh^2 x + \sinh^2 x = \cosh 2x$$

$$(iii) \sinh(-x) = -\sinh x$$

$$\cosh(-x) = \cosh x$$

$$(iv) \sinh 2x = 2 \sinh x \cosh x$$

$$(v) \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$(vi) \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

Derivative $\sinh x, \cosh x, \tanh x$

Other $\operatorname{sech} x, \operatorname{cosech} x, \coth x$

$$\# \log_a x = \frac{\log x}{\log a}$$

$$\# \log_a x = \frac{1}{\log_x a}$$

$$\# \int \ln x dx = x \ln x - x + c.$$

Standard integrals (I)

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C \quad \star \text{ (often confused)}$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log\left(\frac{a+x}{a-x}\right) + C$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log\left(\frac{x-a}{x+a}\right) + C \quad \text{sign}$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C \quad \star \text{ (often confused)}$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log(x + \sqrt{x^2 + a^2}) + C = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad \text{sq. uala x}$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log(x + \sqrt{x^2 - a^2}) + C = \cosh^{-1}\left(\frac{x}{a}\right) + C$$

Integral of type $\int \frac{dx}{A\sqrt{B}}$ where A and B are linear or quadratic function of x.

(i) If B $\rightarrow$ linear, substitution; $B = z^2$

(ii) A linear & B quadratic, $A = 1/z$

(iii) A and B both quadratic, $x = 1/t$

Standard integrals (II)

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log(x + \sqrt{x^2 + a^2}) + C$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log(x + \sqrt{x^2 - a^2}) + C$$

$\rightarrow$ a^2 agadi ko sign

$$\int \sqrt{a^2 - x^2} = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C = -\frac{1}{a} \operatorname{cosec}^{-1}\left(\frac{x}{a}\right) + C$$

$$\int e^{ax} [f(x) + \frac{f'(x)}{a}] dx = \frac{e^{ax} f(x)}{a} + C$$

$\rightarrow \int e^x (f(x) + f'(x)) = e^x f(x) + C$

(ii) $\int [x f'(x) + f(x)] dx = x f(x) + C$

For type $\int (mx + c)\sqrt{ax^2 + bx + d} dx$
Put: linear function = p (Derivative of quadratic) + q

$\frac{d}{dx} \{e^{ax} f(x)\} = ae^{ax} f(x) + e^{ax} f'(x)$
 $\frac{d}{dx} (x f(x)) + C = x f'(x) + f(x)$

$\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C \quad \{n \neq -1\}$

(i) $\int \frac{f'(x) dx}{f(x)} = \ln f(x) + C \quad \{n = -1\}$

(ii) $\int \frac{f'(x) dx}{\sqrt{f(x)}} = 2\sqrt{f(x)} + C$

Integrals reducible to standard forms. (When deno can't be factorized)

(i) $\int \frac{dx}{ax^2 + bx + c}$ (ii) $\int \frac{dx}{\sqrt{ax^2 + bx + c}}$ } only Quadr rat not per for

$\rightarrow$ Make coeff. of x^2 unity by taking coeff. of x^2 outside the quadratic. Then integrand is converted to SI (I), (ii) { & make perfect sq. term inside }

(2) Linear & Quadratic

(i) $\int \frac{mx + c}{ax^2 + bx + c} dx$ (ii) $\int \frac{mx + c}{\sqrt{ax^2 + bx + c}} dx$

Put: Numerator

= p (Derivative of quadratic) + q

Don't get confuse in a^2 &!

Caution while Simplification

+ & -

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \frac{1}{2R} = \frac{2\Delta}{abc}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{iff } \angle A = 60^\circ \Rightarrow b^2 + c^2 - a^2 = bc$$

$$\text{Projection Formula: } a = b \cos C + c \cos B$$

Used to find side of Δ when other two sides & their opp. angle are given.

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\begin{cases} \frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C} \Rightarrow \text{eq. triangle} \\ \cos^2 A + \cos^2 B + \cos^2 C = 1 \Rightarrow \text{Rt. angled triangle} \end{cases}$$

$$\left. \begin{aligned} \sin \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{bc}} \\ \cos \frac{A}{2} &= \sqrt{\frac{s(s-a)}{bc}} \end{aligned} \right\} \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{\Delta}{s(s-a)}$$

$$\therefore \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\rightarrow \Delta = \frac{1}{2} bc \sin A \Rightarrow \sin A = \frac{2\Delta}{bc}$$

$$\Delta = \frac{1}{2} \frac{a^2 \sin B \sin C}{\sin(B+C)}$$

$R \rightarrow$ Circumradius {Rad. of circum circle of triangle}

$$\frac{1}{2R} = \frac{2\Delta}{abc} \Rightarrow R = \frac{abc}{4\Delta}$$

r : inradius {Rad. of incircle of triangle}

$$r = \frac{\Delta}{s}$$

$$r = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}$$

$$r = \frac{a \sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = \frac{b \sin \frac{A}{2} \sin \frac{C}{2}}{\cos \frac{B}{2}} = \frac{c \sin \frac{A}{2} \sin \frac{B}{2}}{\cos \frac{C}{2}}$$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

r_1, r_2, r_3 : Radii of ex-circled circles of a triangle.

$$r_1 = \frac{\Delta}{s-a}; r_2 = \frac{\Delta}{s-b}; r_3 = \frac{\Delta}{s-c}$$

$$r_1 = s \tan \frac{A}{2}$$

$$r_1 = 4R \sin \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}; r_2 = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}; r_3 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

$$A+B+C = \pi \text{ or } 180^\circ$$

$$\text{In any triangle, } \cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

Dimension:

$$\begin{aligned} R, r, r_1, r_2, s &\rightarrow L \\ \sin A, \cos A &\rightarrow 1 \\ \Delta &\rightarrow L^2 \end{aligned}$$

$$\text{In equilateral triangle, } r:R:r_1 = 1:2:3$$

$$\text{In equilateral } \Delta, r_1 = r_2 = r_3$$

Assumption: equilateral triangle of side 2 unit
Rt. angled triangle of sides 3, 4, 5

② → Principle solution (α)
 $\sin \theta \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\cos \theta \rightarrow [0, \pi]$; $\tan \theta \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$

→ General Solution:-

I. $\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$
 $\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$
 $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$

II. $\sin^2 \theta = \sin^2 \alpha$
 $\cos^2 \theta = \cos^2 \alpha$
 $\tan^2 \theta = \tan^2 \alpha$ } $\Rightarrow \theta = n\pi \pm \alpha$

III. $\sin \theta = 0 = \sin 0 \Rightarrow \theta = n\pi$
 $\cos \theta = 0 = \cos \frac{\pi}{2} \Rightarrow \theta = (2n+1)\frac{\pi}{2}$
 $\tan \theta = 0 \Rightarrow \theta = n\pi$

IV. $\sin \theta = -\sin \alpha \Rightarrow \theta = n\pi + (-1)^n (-\alpha)$
 $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + (-\alpha)$

$\cos \theta = -\cos \alpha \Rightarrow \theta = 2n\pi + (\pi - \alpha)$

$\sin \theta = 1 \Rightarrow \theta = (4n+1)\frac{\pi}{2}$
 $\sin \theta = -1 \Rightarrow \theta = (4n-1)\frac{\pi}{2}$
 $\cos \theta = 1 \Rightarrow \theta = 2n\pi$
 $\cos \theta = -1 \Rightarrow \theta = (2n+1)\pi$

→ Squaring while finding roots $\Rightarrow$ check the solutions to satisfy given equation.

→ For $a \cos \theta + b \sin \theta = c$

• Divide by $\sqrt{a^2 + b^2}$

$|c| > \sqrt{a^2 + b^2}$, no solution

$|c| \leq \sqrt{a^2 + b^2}$, has solution.

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

→ Given two variables & asked to find P.V & General value then.
Principle value (PV) of θ : b/w 0 & 2π .

If x lies in

1st quad, PV = x

2nd quad, PV = $\pi - x$

3rd quad, PV = $\pi + x$

4th quad, PV = $2\pi - x$

& General value = $2n\pi + \text{PV}$.

→ Convert to $\sin$ & $\cos$ if question is complex.

→ $\sin^{-1} x \rightarrow$ Angle ; $x = \sin \theta \rightarrow$ Real number.

ITF

$f(x)$	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\operatorname{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$

$\sin^{-1}(-x) = -\sin^{-1} x$; $\cos^{-1}(-x) = \pi - \cos^{-1} x$
 $\tan^{-1}(-x) = -\tan^{-1} x$; $\sec^{-1}(-x) = \pi - \sec^{-1} x$
 $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$; $\cot^{-1}(-x) = \pi - \cot^{-1} x$.

$\sin^{-1} x + \cos^{-1} x = \pi/2$

$\tan^{-1} x + \cot^{-1} x = \pi/2$

$\sec^{-1} x + \operatorname{cosec}^{-1} x = \pi/2$

$2 \sin^{-1} x = \sin^{-1} (2x\sqrt{1-x^2})$

$2 \cos^{-1} x = \cos^{-1} (2x^2 - 1)$

$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$

→ Assumption: suitable value belonging to domain.

→ 'HAT' method: Using options to check answer.

→ Cast Rule

→ $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

→ $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$

→ $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$ $\left\{ \begin{array}{l} \tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \end{array} \right.$

(sine or cosine ko formula main $\because 1 + \tan^2 A = \sec^2 A$)

→ $\sin 3A = 3 \sin A - 4 \sin^3 A$

→ $\cos 3A = 4 \cos^3 A - 3 \cos A$

→ $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$ $\left\{ \begin{array}{l} \text{Sum of One at a time} - \text{Sum of three at a time} \\ 1 - \{ \text{Sum of two at a time} \} \end{array} \right.$

→ $\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cdot \cos\left(\frac{A-B}{2}\right)$

→ $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cdot \cos\left(\frac{A-B}{2}\right)$

→ $\cos A - \cos B = 2 \sin\left(\frac{A+B}{2}\right) \cdot \sin\left(\frac{B-A}{2}\right)$

→ $\sin(A+B) \cdot \sin(A-B) = \sin^2 A - \sin^2 B$

→ $\cos(A+B) \cdot \cos(A-B) = \cos^2 A - \sin^2 B$

→ If $A+B+C = \pi$ (or 180°)

• $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \cdot \sin B \cdot \sin C$

• $\cos 2A + \cos 2B + \cos 2C = 1 - 4 \cos A \cos B \cos C$

• $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}$

• $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

• $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

• $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

→ Domain and Range of $T(f)$

$f(\theta)$	Domain	Range	$f(\theta)$	Domain	Range
$\sin \theta$	$\mathbb{R}$ i.e. $(-\infty, \infty)$	$[-1, 1]$	$\cos \theta$	$\mathbb{R}$	$[-1, 1]$
$\tan \theta$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$\mathbb{R}$ $(-\infty, \infty)$	$\sec \theta$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$
$\csc \theta$	$\mathbb{R} - n\pi$	$(-\infty, -1] \cup [1, \infty)$	$\cot \theta$	$\mathbb{R} - n\pi$	$\mathbb{R}$

→ $c + \sqrt{a^2 + b^2} \geq a \sin \theta + b \cos \theta \geq c - \sqrt{a^2 + b^2}$

• $\sin^2 \theta + \csc^2 \theta \geq 2$

• $\cos^2 \theta + \sec^2 \theta \geq 2$

• $\tan^2 \theta + \cot^2 \theta \geq 2$

→ If $Z = a \sin \theta + b \cos \theta$ such that $a^2 + b^2 = 1$, then min. value of Z is obtained as A.M. $\geq$ G.M.

Quadrant → '+' or '-'

★ Assuming & Getting values

★ Differentiation

★ $\sin x \leq 1$

If difficult to observe anything in question use this

Reverse $(-) \rightarrow (+)$
 $\sin \rightarrow \cos A$

• If $\sec \theta \pm \tan \theta = a$,
 $\sec \theta \mp \tan \theta = \frac{1}{a}$
• If $\csc \theta \pm \cot \theta = a$,
 $\csc \theta \mp \cot \theta = \frac{1}{a}$

$\sec^2 \theta - \tan^2 \theta = 1$

Complex Number

→ $z = a + ib = (a, b)$ $\left\{ \begin{array}{l} x\text{-axis} \rightarrow \text{real axis} \\ y\text{-axis} \rightarrow \text{im-axis} \end{array} \right\}$

→ Every real number is complex number (with $\text{imag part} = 0$)

→ If $z = (a, b)$ & $w = (c, d)$, Then

* $z = w$ iff $a = c$ and $b = d$.

* $z + w = (a + c, b + d)$

* $z - w = (a - c, b - d)$

↳ rep. as diagonal of $\parallel^m$

* $z \cdot w = (a, b) \cdot (c, d)$

$z \cdot w = (ac - bd, ad + bc)$

* $\frac{z}{w} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \left(\frac{ac+bd}{c^2+d^2} \right) + i \left(\frac{bc-ad}{c^2+d^2} \right)$

→ Additive inverse of $z = (a, b)$ is $-z$ i.e. $-z = (-a, -b)$

→ Additive identity is $(0, 0)$

→ For all complex number.

→ Multiplicative identity is $(1, 0)$

→ Multiplicative inverse of $z = (a, b)$ is $1/z$.

$\frac{1}{z} = \frac{1}{a+ib} = \frac{a-ib}{a^2+b^2} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2} \right)$

→ Conjugate of $z = a+ib$ is $\bar{z}$ i.e. $\bar{z} = a-ib$.

→ $z = a+ib$ $\left\{ \begin{array}{l} \rightarrow \text{Purely real if } b=0 \rightarrow \boxed{z = \bar{z}} \\ \rightarrow \text{Purely imaginary if } a=0 \rightarrow \boxed{z = -\bar{z}} \end{array} \right\}$

→ i : imaginary unit of complex number

• $i = (0, 1)$

• $i^2 = (0, 1) \cdot (0, 1) = (0, -1) = -1$

• $i^{4n} = 1$ } where n is an integer

• $i^{4n+2} = -1$

• $\frac{1}{i} = \frac{1}{i} \times \frac{i}{i} = -i$

• Sum of any four consecutive powers of i is zero.

→ Two complex numbers follows associative & distributive law of addition & multiplication and commutative law of addition ~~but not of multiplication~~

→ Properties of complex conjugate:

(i) $(\bar{\bar{z}}) = z$

(ii) $z + \bar{z} = 2 \operatorname{Re}(z)$

(iii) $z - \bar{z} = 2i \operatorname{Im}(z)$

(iv) $\overline{z+w} = \bar{z} + \bar{w}$

(v) $\overline{zw} = \bar{z} \cdot \bar{w}$

(vi) $\overline{z_1 z_2 z_3} = \bar{z}_1 \cdot \bar{z}_2 \cdot \bar{z}_3$

(vii) $\overline{z_1 + z_2 + z_3} = \bar{z}_1 + \bar{z}_2 + \bar{z}_3$

(viii) $\left(\frac{\bar{z}}{w} \right) = \frac{\bar{z}}{\bar{w}}$

★ conjugate gets distributed to each complex number.

Modulus of complex number: $z = a + ib$

$$|z| = \sqrt{a^2 + b^2}$$

- $|z_1| \rightarrow$ distance of z_1 from origin
- $|z_2 - z_1| \rightarrow$ dist. b/w z_1 & z_2
i.e. dist. of z_2 from z_1

$$(i) |z| = |\bar{z}| = |-z|$$

$$(ii) |z| = 0 \text{ iff } z = 0$$

$$(iii) -|z_1| \leq \operatorname{Re}(z_1) \leq |z_1|$$

$$(iv) -|z_1| \leq \operatorname{Im}(z_1) \leq |z_1|$$

$$(v) |z_1 z_2| = |z_1| \cdot |z_2|$$

$$(vi) |z^n| = |z|^n$$

$$(vii) \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0)$$

$$(i)^i \rightarrow \text{Real no.} \rightarrow e^{-\pi/2}$$

$$(viii) |z+w| \leq |z| + |w| \quad \text{Triangle law of inequality}$$

$$(ix) |z-w| \geq |z| - |w|$$

$$(x) |z|^2 = z \cdot \bar{z}$$

$$(xi) |z| = a \Rightarrow \bar{z} = a^2/z$$

$$(xii) |z-w| \leq |z| + |w|$$

$$(xiii) |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

$$(xiv) |z_1 + z_2| = |z_1 - z_2| \Leftrightarrow \arg(z_1) - \arg(z_2) = \pi/2$$

$$(xv) |z_1 + z_2| = |z_1| + |z_2| \Leftrightarrow \arg(z_1) = \arg(z_2)$$

Polar Form of complex number

If $z = x + iy$ then, polar form of z is

Cartesian form. $z = r(\cos \theta + i \sin \theta) \rightarrow \text{cis } \theta$

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = y/x \Rightarrow \theta = \tan^{-1}(y/x) \quad \theta \rightarrow \text{argument or amplitude.}$$

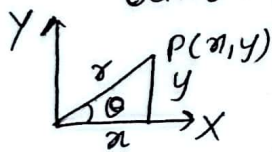


Table for argument: (θ)

- $\theta \in [-\pi, \pi]$
- Argument of $z = 0$ is not defined.

Euler form

$$z = x + iy \rightarrow z = r e^{i\theta} \quad \text{cis } \theta$$

$\rightarrow$ used to find big/bad power of z .
 $e^{i\theta} = \cos \theta + i \sin \theta$

Quadrants	1st	2nd	3rd
Argument	θ	$\pi - \theta$	$-\pi + \theta$
	4th $\rightarrow -\theta$		

For $z = x + iy$
 $\theta = \tan^{-1}(y/x)$

Let, $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$
 $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

Then,

$$* z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$* \frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

De-Moivre's Theorem

For any integer n (positive or negative or zero),

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$$

& the n distinct roots are given by:

$$z_k = \sqrt[n]{r} (\cos \theta_k + i \sin \theta_k)$$

where, $\theta_k = \frac{\theta + k360^\circ}{n}$ & $k = 0, 1, 2, \dots, (n-1)$

Square roots of a complex number; $z = a + ib$

Let, $x + iy = \sqrt{a + ib}$

$(x + iy)^2 = a + ib$

then, $\sqrt{x^2} = \frac{\sqrt{a^2 + b^2} + a}{2}$ and $\sqrt{y^2} = \frac{\sqrt{a^2 + b^2} - a}{2}$

xy and b must have same sign

$\hookrightarrow b = -ve$; take x and y of alternate sign (opposite signs)

$b = +ve$; take x and y of same sign.

Cube roots of unity

$z = \sqrt[3]{1}$

$z^3 - 1 = 0 \Rightarrow (z - 1)(z^2 + z + 1) = 0$

$z = 1$, $z = \frac{-1 \pm \sqrt{3}i}{2}$

$z^2 - z + 1 = 0 \rightarrow$ roots $-w$ & $-w^2$

$z^2 + z + 1 = 0 \rightarrow$ roots w & w^2

$\hookrightarrow 1, \frac{-1 + \sqrt{3}i}{2}, \frac{-1 - \sqrt{3}i}{2}$ i.e. $1, w, w^2$ } $w^2 = \bar{w}$

Properties of cube roots of unity

$1 + w + w^2 = 0 \Rightarrow w + w^2 = -1$

$w \cdot w^2 = w^3 = 1$ } Product of imaginary roots = 1.

$w^{3n} = 1$

• Cube roots of unity forms an eq. triangle in argand plane

• Each im. cube roots of unity is square of other

i.e. $(w)^2 = w^2$ & $(w^2)^2 = w$.

$w^{3k} + w^{3k+1} + w^{3k+2} = 0, k \in \mathbb{Z}$.

Fourth roots of unity $\rightarrow \pm 1, \pm i$

n^{th} roots of unity

$\hookrightarrow$ forms a regular polygon having n sides {eq. Δ , sq. ... }

$\hookrightarrow$ sum of n roots = 0 i.e. $1 + w + w^2 + \dots + w^{n-1} = 0$ } ($n > 1$)

$\hookrightarrow$ Product $\rightarrow (-1)^{n-1}$

Area of Δ formed by z, iz & $z + iz$ as vertices = $\frac{1}{2} |z|^2$

Types of Relation

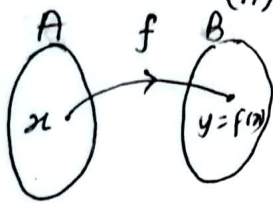
- (i) Reflexive Relation: $[(a, a) \in R, \text{ for every } a \in A] \rightarrow$ Reflexive Relation in set A
 eg: $A = \{1, 2, 3\}; R = \{(1, 1), (2, 2), (3, 3), (1, 3)\} \rightarrow$ Reflexive Relation.
- (ii) Symmetric Relation: $[(x, y) \in R \Rightarrow (y, x) \in R]$
 eg: $A = \{1, 4, 5\}; R = \{(4, 5), (5, 4), (1, 1), (1, 4), (4, 1)\}$
- (iii) Antisymmetric Relation: $[(x, y) \in R \text{ and } (y, x) \in R \Rightarrow x = y]$
- (iv) Transitive Relation: $[(x, y) \in R \text{ and } (y, z) \in R \Rightarrow (x, z) \in R]$
 eg: $A = \{2, 4, 8\}; R = \{(2, 4), (4, 8), (2, 8)\} \because 2 < 4 \text{ and } 4 < 8 \Rightarrow 2 < 8$
- (v) Equivalence Relation: Reflexive, symmetric & transitive

$R \subset A \times B$.

Function

~~From~~ Relation from set A to Set B is said to be function if

- (i) Every element of set A associates
- (ii) To a unique (single) element of set B.



$A \rightarrow$ Domain $\rightarrow x$ which satisfies given $f(x)$
 $B \rightarrow$ Codomain $\rightarrow$ ~~not~~
 $y = f(x) \in B \rightarrow$ Range {Image in B of set A}
 $\rightarrow$ Range $\subset B$ $\hookrightarrow$ {values of y for all possible values of x}

Inequality change $\rightarrow$ (i) Multiplied by '-1'
 (reversed) (ii) Reciprocal

Modulus function $\rightarrow |x| \geq 0$
 $y = |x| \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases}$
 $D_f = R = (-\infty, \infty)$
 $R_f = [0, \infty)$
 $= R - (-\infty, 0)$

Equal function: $f(x) = g(x)$ {for all $x \in A$ }

(i) Domain of $f =$ Domain of $g \Rightarrow D_f = D_g$.

(ii) $f(x) = g(x)$ for all $x \in D_f$.

Number of function from finite set A into a finite set B $\rightarrow \{n(B)\}^{n(A)}$ $\{[T0]$ From

Types of function:-

(i) One to one or Injection

$\hookrightarrow$ If diff. elements in A have diff. images in B.

i.e. For $x_1, x_2 \in A$
 $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$ $\checkmark$

or $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ $\checkmark$

Number of function from a finite set A into a finite set B $\hookrightarrow [n(B)]^{n(A)}$

(ii) Onto or Surjective function.

↳ Every element of B has at least one pre-image in A , otherwise it is said to be into.

i.e. $f(A) = B \rightarrow$ onto functions. { B has pre-image in A }
 $f(A) \subset B \rightarrow$ into function.

✓ Every polynomial function $f: \mathbb{R} \rightarrow \mathbb{R}$ of degree odd is onto & constant with singleton $\rightarrow$ onto function to domain

$\rightarrow$ eg: $f(x) = 3x + 2 \Rightarrow x = \frac{y-2}{3} \in \mathbb{Q} \quad f: \mathbb{Q} \rightarrow \mathbb{Q}$ { $\mathbb{Q}$ - Rationals }
 $f\left(\frac{y-2}{3}\right) = 3\left(\frac{y-2}{3}\right) + 2 = y \Rightarrow y$ is image of x i.e. $\left(\frac{y-2}{3}\right)$
 $\Rightarrow f$ is onto function.

(iii) Bijective function.

↳ Both one to one and one to one i.e. Both injective & surjective.

• Constant function, $f(x) = c$ { singleton }

• Identity function, $y = f(x) = x$

• Linear function, $y = f(x) = mx + c$ { $m, c \rightarrow$ constant & $m \neq 0$ }

✓ Exponential function, $y = f(x) = a^x$, $x \in \mathbb{R}$

$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ $a > 0$ & $a \neq 1$

$= \lim_{h \rightarrow 0} (1+h)^{\frac{1}{h}}$

• Polynomial function, $y = f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$
 $\Downarrow$
 $a_0 \neq 0$ & n is non-negative integer

Polynomial function of degree n

• Logarithmic function: inverse of exponential
 If $y = a^x$, $a \neq 1$ & $a > 0$ its inverse is

$x = a^y$

$\Rightarrow \boxed{y = \log_a x}$

• Reciprocal function: $y = f(x) = \frac{1}{x}$, $x \neq 0$

$D_f = \mathbb{R} - \{0\}$

$R_f = \mathbb{R} - \{0\}$

$xy = 1 \rightarrow$ Rect. hyperbola.

(iv) Inverse function.

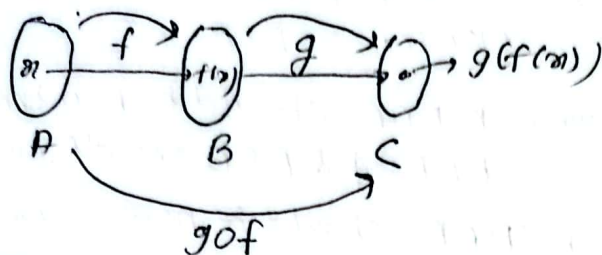
If $f: A \rightarrow B$; then $f^{-1}: B \rightarrow A$ is inverse function of f

Note ① For inverse function to exist; the function must be bijective & i.e. one to one & onto

② If A and B are inverses of each other, then

• Domain of $A =$ Range of B
 • Range of $A =$ Domain of B } Can be used as trick as well.

Composite function
 Let, $f: A \rightarrow B$ and $g: B \rightarrow C$ then composite function of f and g denoted by $g \circ f$ is a function from A to C .
 i.e. $g \circ f: A \rightarrow C$, defined by $(g \circ f)(x) = g\{f(x)\}$ for all $x \in A$



Even function
 $\hookrightarrow f(-x) = f(x)$
 Odd function
 $\hookrightarrow f(-x) = -f(x)$

Periodic function

- period
- (i) $\sin^n x, \cos^n x, \sec^n x, \csc^n x \rightarrow 2\pi \rightarrow n$ is odd
 $\pi \rightarrow n$ is even
 - (ii) $\tan^n x, \cot^n x \rightarrow \pi \rightarrow n$ is even or odd.
 - (iii) $|\sin x|, |\cos x|, |\tan x|, |\sec x|, |\csc x|, |\cot x| \rightarrow \pi$
 - (iv) $|\sin x| + |\cos x|, |\tan x| + |\cot x|, |\sec x| + |\csc x| \rightarrow \pi/2$
 - (v) $f(x)$ is periodic with period $\rightarrow T$, then $f(ax+b)$ is periodic with period $\frac{T}{|a|}$
 - (vi) If $f(x), g(x), h(x)$ are periodic with period T_1, T_2, T_3 then period of $a f(x) \pm b g(x) \pm c h(x)$ is $\hookrightarrow \text{L.C.M. of } \{T_1, T_2, T_3\}$ {where a, b, c are constant}

Note:

$$\text{LCM of } \frac{a}{a_1}, \frac{b}{b_1}, \frac{c}{c_1} = \frac{\text{LCM of } \{a, b, c\}}{\text{H.C.F. of } \{a_1, b_1, c_1\}}$$

Period $\rightarrow$ can be checked from options as well

$A \times B \rightarrow$ Cartesian product

If A contains m elements and B contains n elements then number of elements in $A \times B$ is mn

NO. of possible relation from set A to set $B = 2^{mn}$ {Number of subsets-
 $n(A)$

NO. of functions from finite set A to finite set $B = [n(B)]^{n(A)}$

If $0(A \cap B) = n$, then $0((A \times B) \cap (B \times A)) = n^2$

$|a| < a \Rightarrow -a < x < a$
 $\phi \rightarrow$ empty set { $\{0\} \rightarrow$ set containing only one element i.e. it is singleton set
 (finite set)

Power set of A : set of all subsets of A if $n(A) = n$.

- $\hookrightarrow$ NO. of subsets = 2^n subset in $P(A) = 2^n$
- $\hookrightarrow$ NO. of proper subsets = $2^n - 1$ {except A }
- $\hookrightarrow$ NO. of non-empty proper subset = $2^n - 2$ {except A & ϕ }

Proper subset: If A is proper subset of B if
 $A \subset B$ and $A \neq B$

Equivalent set $\rightarrow$ same no. of elements.

$A \Delta B = (A - B) \cup (B - A)$ } set of non-common
 $= (A \cup B) - (A \cap B)$ } elements of $A \Delta B$.

For $n(A \cup B)$ to be maximum, $n(A \cap B)$ should be minimum
 $n(A \cup B) \leq n(A) + n(B)$ i.e. $n(A \cap B) = 0$ i.e. $A \cap B = \phi$ & A & B are disjoint.

For $n(A \cup B)$ to be min, $n(A \cap B)$ should be maximum
i.e. $A \subset B$.

Idempotent law: $A \cup A = A$ & $A \cap A = A$

Identity law: $A \cup \phi = A$; $A \cap \phi = \phi$; $A \cup U = U$; $A \cap U = A$

De Morgan's law: $\overline{(A \cup B)} = (\bar{A} \cap \bar{B})$ & $\overline{(A \cap B)} = (\bar{A} \cup \bar{B})$

$A - (B \cap C) = (A - B) \cup (A - C)$

$A - (B \cup C) = (A - B) \cap (A - C)$

$A \cap (B - C) = (A \cap B) - (A \cap C)$

$x \notin B \cup C = x \notin B$ and $x \notin C$

$x \notin B \cap C = x \notin B$ or $x \notin C$

$|x + y| \leq |x| + |y|$; $|x - y| \geq |x| - |y|$; $|x|^2 = x^2$

$A \times B \neq B \times A$; $A \subset B$ then $A \times C \subseteq B \times C$ for any set C .

Relation on $A \Rightarrow$ Relation from A to A .

If $n(A) = n$; Number of relation on $A = 2^{n \cdot n} = 2^{n^2}$

Relation $\subset (A \times B)$

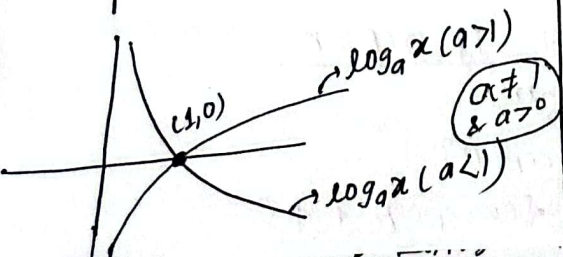
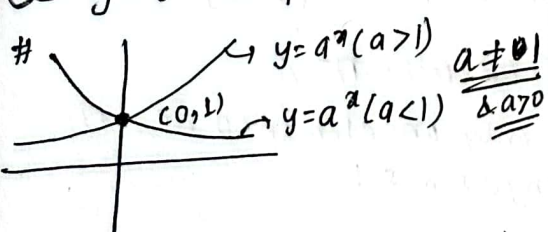
$R = \{(x, y) : x \in A, y \in B\} \subseteq A \times B$

$R^{-1} = \{(y, x) : y \in B, x \in A\} \subseteq B \times A$

$(2^x)^2 = 2^{2x}$; 2^{x^2} if $2^{2x} = 2^{x^2} \Rightarrow 2x = x^2$ or, $x^2 - 2x = 0$

$\sqrt{x^2} = |x|$ or $\pm x$.

$y = e^x \Rightarrow D_f = R$; $R_f(0, \infty)$ } $y = a^x \Rightarrow D_f = R$; $R_f = (0, \infty)$, $a \neq 1$



$\log_a a = 1$, $\log_a 1 = 0$ [$a > 0$, $a \neq 1$]

$\log_a b = \frac{1}{\log_b a}$ [base > 0 & $\neq 1$]

$\log_a b = \log_c b - \log_a c = \frac{\log_c b}{\log_c a}$

$\log_a (xy) = \log_a x + \log_a y$
 $\log_a (x/y) = \log_a x - \log_a y$ } $x > 0$,
 $y > 0$,
base > 0 & $\neq 1$

$\log_a (x^n) = \frac{1}{n} \log_a x$
 $\log_a (x^m) = \frac{m}{n} \log_a x$

Binomial Theorem

$$n! = n \times (n-1)!$$

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y^1 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_n y^n$$

$n \rightarrow +ve$ integer value (Natural number)

$${}^nC_r \rightarrow \text{Binomial coefficient: } {}^nC_r = \frac{n!}{(n-r)! r!}$$

$$\begin{cases} 0! = 1 \\ 1! = 1 \end{cases}$$

* ${}^5C_2 = \frac{5 \times 4}{2 \times 1}$; ${}^6C_3 = \frac{6 \times 5 \times 4}{3!}$ upto 3 numbers (lower index) $\rightarrow$ factorial of lower index.

• ${}^nC_r = {}^nC_{n-r}$, ${}^nC_1 = n$, ${}^nC_n = 1$, ${}^nC_0 = 1$

• General term of $(x+y)^n$

$$T_{r+1} = {}^nC_r x^{n-r} y^r \quad \begin{cases} r \rightarrow \text{integer} ; r \neq \text{negative \& fraction} \\ r \leq n \end{cases}$$

$r \rightarrow (r+1)^{\text{th}}$ term eg: $r=5 \rightarrow 6^{\text{th}}$ term
 $r=4 \rightarrow 5^{\text{th}}$ term

• In expansion of $(x+y)^n \rightarrow (n+1)^{\text{th}}$ term

p^{th} term from end $\Rightarrow (n-p+2)^{\text{th}}$ term from Beginning

• If coefficient of T_r, T_{r+1}, T_{r+2} are in A.P then

$$[(n-2r)^2 = n+2]$$

• In G.P : $S_n = \frac{a(r^n - 1)}{r - 1}$ } $n = \text{number of terms}$

① $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + \infty$
 ② $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + \infty$

• Middle term: $(x+y)^n$

$\rightarrow n \rightarrow \text{even} \rightarrow \text{odd terms} \rightarrow \text{only 1 middle term}$

Middle term: $T_{(\frac{n}{2} + 1)}$

$\rightarrow n \rightarrow \text{odd} \rightarrow \text{even terms} \rightarrow \text{Two middle terms}$

Middle terms: $T_{(\frac{n+1}{2})}, T_{(\frac{n+1}{2} + 1)}$

(8) Coefficient of x^5 in the expansion of $(1+x)^3(1-x^2)^7$

Ⓐ ${}^7C_5 - {}^5C_3$ Ⓑ ${}^{14}C_5$ Ⓒ ${}^3C_1 {}^7C_2 - {}^3C_3 {}^7C_1$ Ⓓ ${}^7C_5 {}^5C_1 - {}^7C_2 {}^5C_3$

Greatest Binomial coefficient Numerically Greatest coefficient

Numerically greatest term (for given value of variable)

→ Greatest Binomial coefficient $\Rightarrow$ Binomial coeff. of middle terms.

• $(1+x)^n \rightarrow$ Greatest Binomial coefficient } same.
Numerically Greatest term

$(x+y)^n$: $m = \frac{n+1}{1 + \left| \frac{x}{y} \right|} \rightarrow$ Numerically greatest coefficient
($x=y=1$)

$n \rightarrow$ index $m-1 \leq r \leq m$ $\left\{ \begin{array}{l} \text{if } m \rightarrow \text{decimal} \\ r \rightarrow \text{integral value b/w} \\ \text{decimal values} \end{array} \right\}$

$m \rightarrow$ position of Numerically Greatest ~~term~~ coefficient

No. of terms in the expansion

• $(x+y)^n \rightarrow (n+1)$ term

$(x+y+z)^n \rightarrow \boxed{n+r-1 C_{r-1}}$ $\left\{ \begin{array}{l} r = \text{no. of variables} \\ \text{here: } x, y, z \Rightarrow r=3 \end{array} \right\}$

Polynomial in single variable having same sign

Eg: $(4x^3 + 5x^2 + x + 1)^6$ $\left\{ \begin{array}{l} \text{variable} \rightarrow x \\ \text{sign} \rightarrow + \end{array} \right.$

$\boxed{\text{No. of terms} = N = (\text{Higher degree} - \text{Lower degree}) \times \text{Index} + 1}$

$N = (3-0) \times 6 + 1 = 19 \text{ terms.}$

Sum of coefficients: put variables = 1.

Eg: sum of coeff. in exp $(ax+by)^n$

sum of coeff = $(a+b)^n$ For $x=y=1$.

$\boxed{(x+y)^n = \sum_{r=0}^n {}^n C_r x^{n-r} y^r}$

$(1+x)^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots + C_n x^n$
 $(1-x)^n = C_0 - C_1 x + C_2 x^2 - C_3 x^3 + \dots + (-1)^n C_n x^n$

Sum of binomial coefficients: $\boxed{C_0 + C_1 + C_2 + C_3 + \dots + C_n = 2^n}$

sum of even & odd coefficients: $\boxed{C_0 + C_2 + C_4 + C_6 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}}$

$$(1+x)^n$$

When, n is not natural number & n is fractional or negative

$$\& \boxed{|x| < 1}$$

↳ coeff. can't be expressed as $nC_0, nC_1, \dots$

$$(1+x)^n = 1 + \frac{n}{1}x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \infty \text{ terms}$$

$$|x| < a \Rightarrow -a < x < a$$

g) Coefficient of x^3 in expansion $\frac{(1+3x)^2}{1-2x}$

- (i) 50 (ii) 32 (iii) 48 (iv) 55

≠ Sum of Product of Binomial coefficients $\{C_0C_x + C_1C_{x+1} + \dots\}$
 $= \frac{(2n)!}{(n-x)!(n+x)!}$ } If difference of lower index is constant
 $\cdot x \rightarrow$ difference of lower indices

eg: $C_0C_2 + C_1C_3 + C_2C_4 + \dots + C_{n-2}C_n$
 $= \frac{(2n)!}{(n-2)!(n+2)!}$

★ Putting value of n to find sum
 If given other constant in sum like a, b then take $n \geq 2$

$a_0C_0 + a_1C_1 + a_2C_2 + a_3C_3 + \dots + a_nC_n$
 If a_0, a_1, a_2, a_3 are in A.P., then
 $= \frac{(2a_0 + nd)2^{n-1}}{2}$ } $a_0 \rightarrow$ first term
 $d \rightarrow$ common difference.
or $S = (a_0 + a_n)2^{n-1}$

$\frac{(x+a)^n + (x-a)^n}{2}$
 Total number of terms $= \left(\frac{n}{2} + 1\right)$ if n is even
 (After simplification) & $\left(\frac{n+1}{2}\right)$ if n is odd

$\frac{(x+a)^n - (x-a)^n}{2}$
 Total number of terms in the expansion after simplification
 $= \frac{n}{2}$ if n is even
 $= \frac{n+1}{2}$ if n is odd

In expansion $\left(x^p + \frac{1}{x^q}\right)^n$
 For term independent of x , } $x = \frac{n \times p}{p+q}$

Ans: $C(n, x)$
 $x \rightarrow$ -ve or fraction
 No term independent of x .

→ For $(x^p - \frac{1}{x^q})^n$: Independent term of x

x can be found in some way for other question of x^n but be caution while finding coeff.

$x = \frac{n \times p}{p+q}$ } Ans: $C(n, x)$ if x is even
 $-C(n, x)$ if x is odd.

(9) The independent term in expansion

$$\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - \sqrt{x}} \right)^{10} \text{ is}$$

$\frac{1}{(-ve)!} \Rightarrow$ considered 0.

- ① $C(10, 0)$ ② $C(10, 3)$ ③ $C(10, 10)$ ④ $C(10, 4)$

$(-ve)! =$ not defined
 $e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = \lim_{n \rightarrow 0} (1+n)^{\frac{x}{n}}$

For $(ax^p + \frac{b}{x^q})^n$

no terms in x innumerable for finding.

To find coefficient of x^m in expansion

$x = \frac{n \times p \times -m}{p+q}$

$\frac{\ln AP}{S_n} = \frac{n}{2} [2a + (n-1)d]$

Next, use above x to find coefficient.

$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \{ e^{a+b} = e^a \cdot e^b \}$

$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \{ \text{For } -1 < x \leq 1 \}$

$a^x = 1 + \frac{x}{1!} \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \dots \rightarrow a > 0. \{ \text{factorial} \rightarrow ! \Rightarrow \text{In exp. of } e^x \}$

$\log_e(1-x) = -\frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \{ \text{For } -1 \leq x < 1 \}$

The coefficient of x^n in the expansion $\log(1+x+x^2)$ is

- $-2/n$ if n is multiple of 3
- $1/n$ if n is not multiple of 3

$\log a - \log b = \log \left(\frac{a}{b}\right)$ # $\log a + \log b = \log(ab)$

In expansion $(x+1)(x+2)(x+3) \dots (x+n)$

coeff. of $x^{n-1} = \frac{n(n+1)}{2}$

$\log_e \left(\frac{1+x}{1-x}\right) = 2 \left(\frac{x}{1} + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right)$
 for $-1 < x < 1$ i.e. $|x| < 1$.

If the coeff. of p^{th} and q^{th} terms in the expansion of $(1+x)^n$ are equal, then

$p+q = n+2$

In GP
 $S_n = \frac{a(x^n - 1)}{x - 1}$

$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots \infty$ } only even terms
 (don't forget $1/n!$)